Dynamic Surveys

AI Proof of Concept Scope Document

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Industry: Public Option and Survey Research
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Opportunity Assessment

	Description
Problem	Traditional survey instruments rely on static, rule-based logic flows (skip patterns) that fail to adapt to a respondent's context or engagement level. Consequently, open-ended questions—the primary tool for capturing qualitative nuance—suffer from high nonresponse rates, reaching up to 25%. This participation gap is not distributed evenly; younger, Black, and Hispanic respondents are significantly less likely to provide detailed written feedback, creating an "Equity Trap" that skews data representation. Furthermore, mobile respondents typically provide answers that are 30% shorter than those from desktop users. This lack of depth undermines methodological rigor, creates vague low quality data and drives up the cost by requiring subsequent qualitative research such as human follow up through In-depth interveiws and Focus Group Discussions. (Barari et al., 2024; Gómez & Amina, 2023)
Opportunity	Agentic Probing Bots: Generative AI systems designed to transform static questionnaires into interactive, conversational experiences. By utilizing real-time semantic reasoning, the system can identify vague, one-word, or incomplete responses. Rather than moving to the next question, the AI acts as a neutral moderator, dynamically generating context-aware follow-up questions to clarify intent and extract deeper insights. To maintain longitudinal consistency, the architecture will utilize Retrieval-Augmented Generation (RAG) to prioritize historic, pre-approved survey questions from a standardized database before generating new survey prompts / questions.

	Description
	I.e LLMs generate dynamic surveys that automatically ask follow-up questions to vague or incomplete responses through the use of an AI chatbot
Expected Benefits	- **Increased response richness (40–60% longer qualitative answers)**: Deploying an Agentic Probing Bot that mimics a neutral human moderator transforms short, vague phrases into comprehensive data. Early industry data demonstrates this increases open-ended character counts by 40–60% without increasing respondent attrition (Barari et al., 2024). - **Reduced nonresponse bias across demographic groups:** actively combats the 25% nonresponse rate typical of open-ended questions by engaging respondents conversationally. It is particularly effective at expanding "thin" mobile answers, which are traditionally 30% shorter than desktop responses (Gómez & Amina, 2023). - **Lower operational cost ($10k–$15k savings per study)**: By delivering focus-group-quality depth at web-survey scales, the dynamic AI eliminates the need for expensive, subsequent human-led in-depth interviews or Focus Group Discussions. This consolidation saves an estimated $10,000 to $15,000 per study, based on standard industry pricing (Merren, 2026). - **Improved respondent engagement and completion rates** (Barari et al., 2024). - **Reduced manual programming burden for IT teams since they now longer have to program/script skip logic**
Strategic Alignment	This AI-driven approach directly supports the core mandate of organizations like the Pew Research Center to deliver the "gold standard" of rigorous, representative public opinion data amid a shifting landscape: ● **Combating Structural Industry Challenges:** As the public moves away from landlines and ignores unknown contacts (McClain, 2020), overall response rates have plummeted—with federal BLS surveys dropping from 65% to nearly 40% (Leduc & Oliveira, 2025). GenAI addresses this participation crisis by

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	making the survey experience inherently dynamic, conversational, and worth the respondent's time ● **Restoring Methodological Equity:** Traditional static surveys suffer from a demographic "Equity Trap," where younger, Black, and Hispanic adults or those responding are significantly less likely to write open-ended responses. Conversational probing specifically rescues the missing "why" from these demographics by further probing in an effort to gain more detailed responses, protecting the institution's reputation for highly representative data. ● **Defending Market Position:** Low-cost online panels and social media "listening" are eroding differentiation while introducing documented coverage and nonresponse biases (Diaz et al., 2016). GenAI neutralizes this threat by reducing the cost disparity between premium, probability-based surveys and cheap opt-in panels, allowing the organization to offer premium-quality depth without the premium price tag associated with more human approaches like In-depth Interviews or Focus Group Discusions.
Success Measures	● **Latency:** The API endpoint generates and returns the follow-up probe in under 1.5 seconds (this is to reduce attrition and drop off - people won't have the patience to wait for a response). ● **Data Quality:** Increases in the open-ended character count by at least 40% compared to a static control survey question ● **Neutrality:** The system passes bias benchmarks and does not ask "leading" questions that skew respondent data. ● **Completion rate increased by 15%** ● **Average response length increased by 40%** ● **Drop-off rate reduced by 20%**
Risk Considerations	● **Latency and Attrition Risk:** Delays in generation time (>1.5s) respondent may become impaitian and drop off, directly driving survey abandonment.

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	● **Fatigue Risk:** If the model's fails to recognize when a response is sufficient, or if the model excessivly probes the respondent (> 2 probes per question) it may probe the respondent excessively, they may become frustrated or annoyed and abandon the survey - increasing drop off ● **Bias Risk:** The model might inadvertently lead the respondent (e.g., assuming a political stance based on a vague answer). ● **Output quality:** Model responses should be structured like a normal survey. There shouldn't be ● **Data privacy and API exposure risks** ● **Regulatory Compliance:** e.g., GDPR or Emerging Regulation like the EU's AI Act ● **Over-automation reducing methodological transparency**

Stakeholder Analysis

Stakeholder	Description	Current Challenges	Potential Benefits
Survey Respondents	The individuals participating in the public opinion polls include hard-to-reach demographics.	Survey fatigue from rigid, static forms and frustration leading to a 25% nonresponse rate on open-ended questions. (Barari et al., 2024; Gómez & Amina, 2023)	A more engaging, conversational survey experience that feels like a natural dialogue, encouraging them to share detailed perspectives.
Survey Methodologists	Researchers are responsible for survey design, data collection, and statistical validity.	Getting high-quality, detailed responses from open-ended questions, non-response biases, as younger, Black, and Hispanic adults are less likely to share written	The AI mimics a neutral human moderator and encourages more detailed responses, including more of the "why" behind the data and increasing character counts by 40-60%. (Barari et al., 2024).

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		views. (Gómez & Amina, 2023).	
Organization Executives	Leadership at research organizations is responsible for strategy and institutional reputation.	Losing clients to low-cost competitors while facing prohibitive costs to maintain high-quality "Gold Standard" surveys.	Delivers premium data depth without the premium price, saving $10,000 to $15,000 per study by eliminating manual human follow-ups. (Merren, 2026).
Engineering & IT Teams	Developers and Machine Learning engineers are managing system architecture and deployment.	Integrating real-time webhooks securely without disrupting the survey and maintaining strict latency to prevent user drop-off.	Replaces Fragile Rule-Based "skip logic" found in traditional survey or basic chatbots that need to be manually hardcoded with exhaustive decision trees, the use of an LLM would reduce the IT team's manual programming and maintenance burden.
Compliance & Ethics (IRBs)	Legal and ethical experts serving on the Institutional Review Board.	Ensuring adherence to global data privacy laws, including GDPR and CCPA (jamesju, 2025; Regulation (EU) 2016/679, 2016; Regulation 2024/1689, 2024) Mitigating algorithmic bias where models might flag non-standard dialects (e.g., AAVE) as "toxic" or ask leading questions (Poole-Dayan et al., 2025)	Helps reduce nonresponse bias, as younger, Black, and Hispanic adults are less likely to share written views in standard open-ended questions. (Gómez & Amina, 2023). Plus automated Privacy Filters redact PII locally before processing to ensure proactive regulatory compliance (Regulation (EU) 2016/679, 2016).

Objectives and Key Results

Objective	Key Results	Metrics
Objective 1: Increase Qualitative Data Depth Without Increasing Attrition	1. Increase open-ended response length by ≥40% compared to standard surveys 2. Reduce open-ended nonresponse rate from 25% to ≤15%. 3. Maintain or improve completion rates (+15%). 4. Maintain average probe latency under 1.5 seconds.	● Average character count ● Open-ended question completion rate ● survey completion rate ● API latency logs
Objective 2: Ensure Equitable Data Collection by Mitigating Demographic Nonresponse Bias	1. Achieve ≥95% classification accuracy in detecting and correctly interpreting vague versus substantive responses across a balanced validation set that includes regional dialects, colloquialisms, and non-standard syntax. 2. Increase average open-ended character count among historically low-response subgroups (e.g., younger, Black, Hispanic respondents) by ≥25% relative to baseline control (static survey condition) 3. Reduce disparity in system-triggered friction	● BLEU, ROUGE, and METEOR scores (Rosario & Noever, 2023) ● Average character count by demographic group ● Friction Event Rate (number of friction events) ● Completion Rate by Subgroup

Objective	Key Results	Metrics
	events (e.g., repetitive probing, false vague-flagging, bias-trigger regeneration, or rejection errors) between minority subgroups and majority subgroup to ≤5% difference from baseline friction rate. 4. Achieve a qualitative survey completion rate where the variance across all targeted demographic subgroups deviates by **no more than 5%** from the overall median survey completion rate.	
Objective 3: Improve Operational Efficiency and reduce the cost of deploying surveys	1. Reduce the manual programming time by >30% for IT teams by replacing fragile, rule-based skip logic with adaptive LLM pathways. 2. Achieve $10,000 to $15,000 in estimated operational savings per deployment by directly eliminating manual human follow-ups. 3. Maintain strict system latency standards when integrating real-time webhooks, ensuring AI response delays do not exceed 10% of the average time a respondent spends answering a survey question, thereby	● IT and engineering hours logged for survey architecture setup ● Cost per completed survey ● Budget comparison versus traditional human-moderated qualitative studies ● Average API inference latency in milliseconds ● Median question Response time

Objective	Key Results	Metrics
	preventing latency-induced drop-offs 4. Accelerate the survey pilot testing and Quality Assurance (QA) timeline from 14 days to under 5 days by eliminating the need to manually verify exhaustive, rule-based branching logic trees.	

Responsible AI Assessment

AI Risk Impact Matrix

Stakeholder Groups	Concerns and Potential Impacts				
	Trust Concerns	**Fairness Concerns**	**Safety Concerns**	**Transparency Concerns**	**Sustainability Concerns**
Direct Users **Researchers + Analysts** *People who directly interact with or use the system*	Undocumented updates to the LLM backend altering probing behavior mid-fielding / in between rounds of the same survey	Automatic Purging or flagging of "low-quality" or "bot-like" responses for removal could penalize responses from those with lower level english proffiency/ education	The AI inadvertently prompting respondents to share Personally Identifiable Information during unscripted follow-ups.	Inability to document the exact survey instrument or prompt chain used for any specific respondent in a methodology report.	High API token expenses when running dynamic inference on massive, nationally representative samples.
Subjects **Survey Respondents** *People who are subject to decisions made by the system*	Respondents feel manipulated if the AI mimics human empathy.	The AI failing to understand or probe responses written in regional dialects, slang, or non-standard grammar.	AI autonomously generating highly inappropriate or invasive follow-up questions on	Lack of informed consent that an LLM is reading and reacting to text in	Repeated exposure to AI-mediated conversational surveys may contribute to

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	Trust Concerns	**Fairness Concerns**	**Safety Concerns**	**Transparency Concerns**	**Sustainability Concerns**
			sensitive or traumatic topics.	real-time vs a human	long-term survey fatigue if interactions feel artificial, repetitive, or cognitively demanding. Over time, this could reduce willingness to participate in probability-based research, particularly among already underrepresented groups.
Organization **Survey Research Organizations** *Entity or entities deploying/operating the system that aligns with their strategic objectives*	Over-reliance on opaque third-party LLM providers may reduce institutional control over methodological integrity, especially if backend model weights, safety filters, or alignment tuning change without advance notice. Sudden behavioral drift could undermine longitudinal comparability across survey waves.	Cross-national deployment may introduce cultural bias if the probing style/norms are optimized for U.S. respondents are applied globally without localized validation.	**Data Breaches:** Passing unanonymized, sensitive microdata through third-party APIs.	Complex AI pipelines (RAG retrieval, bias filters, regeneration loops) may reduce explainability for non-technical stakeholders, creating blind spots unless everything is well documented	**Vendor Lock-in**: Total reliance on proprietary third-party LLMs to field core research instruments.

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	Trust Concerns	**Fairness Concerns**	**Safety Concerns**	**Transparency Concerns**	**Sustainability Concerns**
Domain Experts ***Survey methodologists, statisticians, IRB members, policy analysts*** *Subject matter experts and practitioners*	Concern that AI-generated probes may subtly shift question meaning, compromising construct validity and threatening comparability with historic probability-based instruments.	Fear that model tuning decisions (e.g., temperature, prompt framing) may unintentionally privilege certain ideological frames, altering the interpretability of sensitive political or social data.	Risk that AI probing introduces measurement error by encouraging over-elaboration , socially desirable response Bias, or speculative answers beyond respondents' original intent.	Difficulty in reproducing AI-mediated survey interactions for peer review, replication studies, or academic publication due to dynamic prompt generation.	Dependence on specialized AI engineering expertise may create institutional knowledge gaps if technical staff turnover occurs.
Broader Community *Society and groups indirectly affected*	Total loss of faith in polling if dynamic surveys are perceived as manipulative "push polls" leading respondents to specific conclusions.	Minorities being systematically underrepresented in qualitative summaries because their phrasing was not recognized by the model.	Dynamic surveys that attempt to "match" respondent sentiment might inadvertently validate or amplify extreme political views.	Public misunderstanding about how AI systems shape survey questioning may fuel narratives that polling organizations are "engineering" public opinion rather than measuring it.	The massive environmental energy footprint required to run live inference for millions of individual survey questions.

Risk Level Definitions

- **Serious Concerns (Red)** Critical harm or violation of laws/ethics requiring immediate, high-priority action
- **Moderate Concerns (Orange)** Noticeable impact but manageable risk requiring planned action
- **Minimal Concerns-White/No color.** Negligible impact or cosmetic issues requiring monitoring only

Governance Framework

Governance Area	**Requirements**	**Evaluation Approach**	**Success Criteria**
Policy & Regulation	Must adhere to strict global standards including the EU's GDPR, the CCPA, and the EU AI Act. Must follow AAPOR and ESOMAR ethics guidelines.	Review of the data pipeline against a predefined checklist of data minimization and consent principles prior	Both members complete the compliance checklist individually and the survey includes mandatory opt-in consent before proceeding.
Algorithmic & Instrument Bias	Must prevent AI from generating "leading" questions or penalizing non-standard dialects, such as AAVE, which could silence underrepresented minorities.	Feed a controlled test batch of 20-30 varied, synthetic responses (spanning different dialects and viewpoints) into the model and manually audit the outputs..	No statistically significant differences in probe tone or politeness across key demographic subgroups.
Transparency & Auditability	Must guarantee informed consent, explicitly notifying respondents when they are interacting with an AI agent. Must ensure full traceability of AI logic by	Automated system logging and regular review of the AI override/fallback rates. Visually verify that disclaimers are present on all AI-driven questions.	100% of AI interactions documented; AI hallucination/leading-question error rate remains strictly below 0.5%. All questions that are processed or

Governance Area	Requirements	Evaluation Approach	Success Criteria
	Log all prompts, model outputs, and configuration versions		generated by an LLM has a disclaimer and additional disclaimer at the beginning of the survey
Privacy Measures	PII redactied before processing and Must ensure raw Personally Identifiable Information (PII) is never transmitted to external model APIs.	Local filtering of PII along with Inputing fake PII (e.g., dummy names, phone numbers) into the chat interface during testing to verify the local regex/privacy filter catches it.	100% of injected fake PII is successfully redacted in the local logs *before* sent to th e API.
Security Controls	Encrypted data transfer, access restrictions	Penetration testing + monitoring	No critical vulnerabilities
Data Management	Standardized schema and retention policies	Define a simple JSON schema for the collected data and review the final output file after a test run.	The POC generates a properly formatted JSON/CSV output file with no missing critical data fields.

Unsupported Uses Matrix

Use Case	Rationale for Exclusion	Risk Level
Translation / Adaptation of International Questionnaires	Utilizing LLMs for direct, unverified translation of survey instruments across different languages and cultural contexts risks stripping away vital local idioms and cultural context, altering the fundamental meaning of the question, and compromising cross-national comparability.	Medium
Dynamic offerings or Adjusting Incentive	Using AI to analyze a respondent's hesitation or engagement level (e.g., response latency) and dynamically offering higher financial incentives mid-survey to prevent break-offs. This creates a perverse	High

Use Case	Rationale for Exclusion	Risk Level
	incentive structure and could bias the final sample toward respondents who "game" the system, and violates standard ethical practices regarding equal compensation for equal burden.	
Real-Time Generation of Multiple Choice / Closed-Ended Response Options	Dynamically altering or generating multiple-choice response categories based on earlier respondent answers increases the risk of the model leading/creating a source of bias for respondents (choice of certain options, or non-mutually exclusive options) or it could mess with statistical comparability, e.g., Respondent A receives four options, and Respondent B receives a dynamically generated six	Medium

Application Architecture

Component	RAG Application	Agent Application
Model	*- Embedding-gemma-300m* *- Standard dimensions* *- Vectorizing historical survey codebooks and metadata from Pew and WVS datasets.*	*- Gemma 3 12B* *- Low temperature (e.g., 0.1-0.2) to ensure deterministic, neutral responses.* *- Generating real-time, context-aware follow-up probes to clarify vague open-ended responses.*
Vector Store	*- Astra DB or Chroma DB* *- Cosine similarity search* *- API connected to the Agent application.*	*Structure: N/A (Does not store vectors directly).* *Search: N/A.* *Integration: Queries the RAG vector store via API to retrieve contextually relevant survey phrasing before generating a probe.*
Prompts	***Design:*** *Retrieves pre-tested public opinion questions related to the topic of the question and Respondent response.* ***System Instructions:*** *Prioritize exact matches from the WVS and Pew codebooks over generalized semantic matches.*	***Design:*** *Inject the respondent's prior answer into the prompt context and then probe the respondent (1-2 times max) with follow-up questions* ***System Instructions:*** *"Act as a neutral human survey moderator. Do not lead the respondent. Ask exactly one brief, clarifying follow-up question based on*

Component	RAG Application	Agent Application
		their previous text. Model question on existing question in database"
Governance Controls	***Safety:*** *Validate all retrieved questions originate from the approved, pre-tested datasets.* ***Checks:*** *Automated relevance scoring of retrieved chunks.* ***Monitoring:*** *Audit logs of query-to-retrieval mappings*	***Safety:*** *Real-time, local PII redaction before text hits the LLM API .* ***Checks:*** *Algorithmic bias filters to prevent misinterpretation of non-standard dialects. Another LLM used to check for Bias or Leading Questions* ***Monitoring:*** *Live tracking of webhook latency and token costs.*
Human Oversight	**Review points:** Methodologist review of the top 50 most frequently retrieved historical questions to ensure they are being applied in the correct modern context **Control:** Strict semantic similarity thresholds (e.g., cosine similarity must be > 0.85); if no historical question meets the threshold, the RAG returns an empty context rather than guessing. **Override:** Hard-coding specific Pew/WVS questions to trigger when certain predefined respondent keywords are detected, intentionally bypassing the semantic vector search.	***Review Points:*** *Daily sampling of dynamic transcripts to verify neutral tone.* ***Control:*** *Strict limits on the maximum number of probes per respondent.* ***Override:*** *Automatic fallback to static rule-based questions if the API times out or fails safety checks.*

Component	RAG Application	Agent Application
Performance & Quality Metrics	***Performance:*** *Retrieval latency < 200 milliseconds.* ***Quality:*** *>85% semantic match rate when retrieving relevant historical survey questions*	***Performance:*** *AI response delays do not exceed 10% of the median question response time.* ***Quality:*** *40-60% longer qualitative answers; 15% increase in survey completion rates.*
Dependencies	*High-quality metadata extraction from Pew (.sav/.xml) and WVS datasets. Uninterrupted vector database uptime.*	*Seamless webhook integration with the primary survey hosting platform. RAG application availability for contextual grounding.*

Data Sources

Data Source	Type	Description	Purpose	Preparation Needs
Pew Global Attitudes Survey (Spring 2024, 2014)	Cross-national public opinion microdata (.sav, .csv) and metadata (.xml, data dictionaries)	Archives of past probability-based survey questionnaires and codebooks focusing on global public opinion, economic attitudes, and political views.	Populates the RAG Question Bank to retrieve pre-approved, methodologically sound questions related to current events and policy for consistent follow-up probing.	Extract codebook labels from the Data Dictionary. Data cleaning, semantic chunking, and vectorization to load the historical text into the vector database.
World Values Survey (WVS7)	Cross-national public opinion microdata (.sav)	Archives of questionnaires and codebooks tracking cross-cultural beliefs,	Expands the RAG Question Bank to provide the Agentic Probing Bot with standardized, pre-tested	Extract codebook labels from the Data Dictionary. Data cleaning Alignment of

Data Source	Type	Description	Purpose	Preparation Needs
		values, and social norms over time.	questions for deep qualitative probes on cultural or sociological topics.	coding schemas with the Pew dataset, followed by semantic chunking and embedding generation for the vector database.

Proof of Concept Roadmap

Component	Sprint 1	Sprint 2	Sprint 3
Timeframe	Week 1 - 2	Week 3 - 4	Week 5
Goals	Build a functional prototype foundation	Implement AI probing, integrations	Validate system performance & finalize demo
Use Cases	Static survey, interactive interface	Context-aware probing responses	End-to-end survey simulation
Features	UI mock survey + API connection	Prompt logic, memory context	Bias check, logging dashboard
RAI & Governance Evaluation	Implement explicit informed consent notices and test local PII filtering tools to ensure no sensitive data reaches the API	Conduct audits on probe tone to ensure its neutral across simulated demographics and dialect. Test human-fallback mechanisms	Complete the compliance checklist individually to ensure compliance with GDPR
Success Metrics	Prototype runs locally with 0 API connection failures; Privacy Filter	AI generates probes in under 1.5 seconds; achieves a 20%	15% overall increase in completion rate in pilot run; AI

Component	Sprint 1	Sprint 2	Sprint 3
	catches 100% of injected PII.	increase in response length over the control in mini pilot	hallucination rate strictly below 0.5%.
Key Tasks	Architecture setup, dataset prep	LLM tuning, prompt testing	Evaluation, Reporting, Presentation

Appendix

Functionality and Features

Name	Description	Goal	Definition of Done
Local On-Device Functionality	The model should theoretically be able to run on a mobile or a laptop	To allow surveys to be taken offline and keep sensitive data completely private on the user's device.	The AI can successfully generate a follow-up question without an internet connection and using a Small Language Model
Privacy Filter	A safety tool that automatically detects and removes personal information from the chat.	Protect respondent privacy (like names or SSNs) before data is processed.	The system successfully finds sensitive info and replaces it with [REDACTED].
Smart Question Generator	Reads the user's short answers and generates follow up question	To ask a specific, relevant question based on what the user just said.	The AI successfully asks a logical follow-up question directly related to the user's input.
Identify Vague and incomplete responses	A system that checks if a respondent's answer is too short, unclear, or incomplete.	Model figures out exactly when the AI needs to step in and ask a follow-up question, while ignoring perfectly good, detailed answers.	The system accurately flags 1-3 word or non-descriptive answers (like "I don't know" or "inflation") and lets detailed paragraphs pass without triggering the AI.
Bais Check	Before outputting the response to the user, AI checks to make sure that its response isn't leading but instead is neutral	To prevent any accidentally leading or biased questions from being sent.	If a question is flagged, the system automatically regenerates it
Question Bank RAG	When possible, the system looks up approved historic survey questions from a database and either modifies them, keeps them as is, help the craft a new question.	To reuse standardized questions and keep the survey consistent with past research, and reduce computing time	The AI successfully finds and uses a relevant, pre-approved question from the database instead of inventing one entirely from scratch.

In Scope and Out of Scope Functionalities

In Scope	Out of Scope
● Building the probing logic using Langflow. ● Exposing the completed Langflow pipeline as a REST API endpoint and adding it to a mock survey platform similar to a chat interface. ● Utilizing Langflow Prompt Template and LLM nodes with efficient, fine-tuned smaller models (4B–12B parameters) to prioritize speed and cost. ● Implementing memory nodes in Langflow to track the context of a specific user's survey session.	● Using massive "Frontier" models (like GPT-5) for simple probing (too expensive/slow). ● Audio/Voice-based probing (text interaction only for this PoC). ● Replacing probability-based human sampling entirely (AI is only assisting the human respondent, not replacing them).

Mock Conversation

Before	After
Survey Prompt: "In your own words, what do you think is the most important problem facing the country today?" **Respondent Input:** "The economy and inflation." **System Action:** *No intervention.* The survey immediately advances to the next question. **Methodological Result:** The researcher receives a vague, four-word answer and offering no actionable qualitative nuance	**Survey Prompt:** In your own words, what do you think is the most important problem facing the country today?" **Respondent Input:** The economy and inflation. **System Action:** The Langflow API endpoint evaluates the response length and semantic depth. It flags the response as vague/incomplete. Within 1.5 seconds, the RAG application queries the vector store (Astra DB) for methodologically approved follow-up structures. **Agentic Probe (Generated by LLM):** Thank you for sharing that. You mentioned the economy and inflation. Could you tell us a little more about how that specific issue is impacting your day-to-day life? **Respondent Input (Follow-up):** Groceries are just way too expensive right now. I feel like I'm spending $50 more a week on the exact same basic items like eggs and milk compared to last year. It makes it really stressful to try and save any money for emergencies when the basics cost so much. **System Action:** The system successfully evaluates the response as substantive and advances to the next survey section. **Result:** The open-ended character count increases dramatically The researcher captures the 'why'" , transforming a generic response into a rich, detailed data point without requiring an expensive human-led follow-up interview.

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